

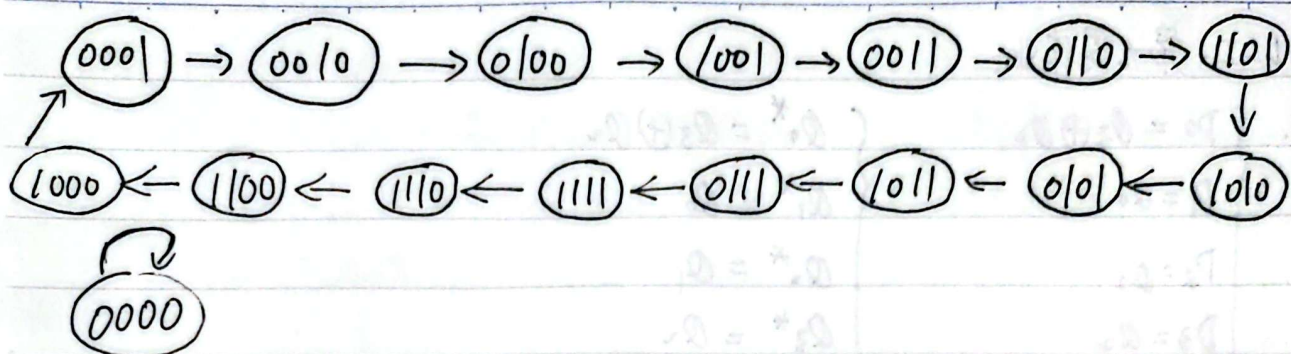
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VLSI - 第九周作业.

$$1. \begin{cases} D_0 = Q_3 \oplus Q_2 \\ D_1 = Q_0 \\ D_2 = Q_1 \\ D_3 = Q_2 \end{cases}$$

$$\begin{cases} Q_0^* = Q_3 \oplus Q_2 \\ Q_1^* = Q_0 \\ Q_2^* = Q_1 \\ Q_3^* = Q_2 \end{cases}$$

Q_3^*	Q_2^*	Q_1^*	Q_0^*	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0	0	0	0
✓ 0	0	1	0	0	0	0	1
✓ 0	1	0	0	0	0	1	0
✓ 1	0	0	1	0	1	0	0
✓ 0	0	0	1	1	0	0	0
✓ 0	1	1	0	0	0	1	1
✓ 1	0	1	1	0	1	0	1
✓ 0	0	1	1	1	0	0	1
✓ 1	1	0	1	0	1	1	0
✓ 0	1	0	1	1	0	1	0
✓ 1	0	0	0	1	1	0	0
✓ 1	1	1	1	0	1	1	1
✓ 0	1	1	1	1	0	1	1
✓ 1	0	1	0	1	1	0	1
✓ 1	1	0	0	1	1	1	0
✓ 1	1	1	0	1	1	1	1



综上所述，这是+5进制计数器。不能自启动。

2. 74LS161. 异步清零 同步置数. = 82.

两片之间

→ 即 $5 \times 16 + 2 = 82$

当 $Q_3^{(2)} Q_2^{(2)} Q_1^{(2)} Q_0^{(2)} Q_3^{(1)} Q_2^{(1)} Q_1^{(1)} Q_0^{(1)} = 0101 0010$ 时 同步置数.

初始置数为 0000 0000 即 "0"

计数从 0 - 82 且 82 有效

综上所述 此为 83 进制计数器 两片之间是 16 进制.

3. A

R	Y	G	Q_2	Q_1	Q_0
0	0	0	0	0	0
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	0	1	1
1	1	1	1	0	0
0	0	1	1	0	1
0	1	0	1	1	0
1	0	0	1	1	1
0	0	0	0	0	0

$$\therefore R = Q_2' Q_1' Q_0 + Q_2 Q_1' Q_0' + Q_2 Q_1 Q_0$$

$$= 1 + 4 + 7$$

$$Y = Q_2' Q_1 Q_0' + Q_2 Q_1' Q_0' + Q_2 Q_1 Q_0'$$

$$= 2 + 4 + 6$$

$$G = Q_2' Q_1 Q_0 + Q_2 Q_1' Q_0 + Q_2 Q_1' Q_0$$

$$= 3 + 4 + 5$$

138 算中规模吗. 算吧.

